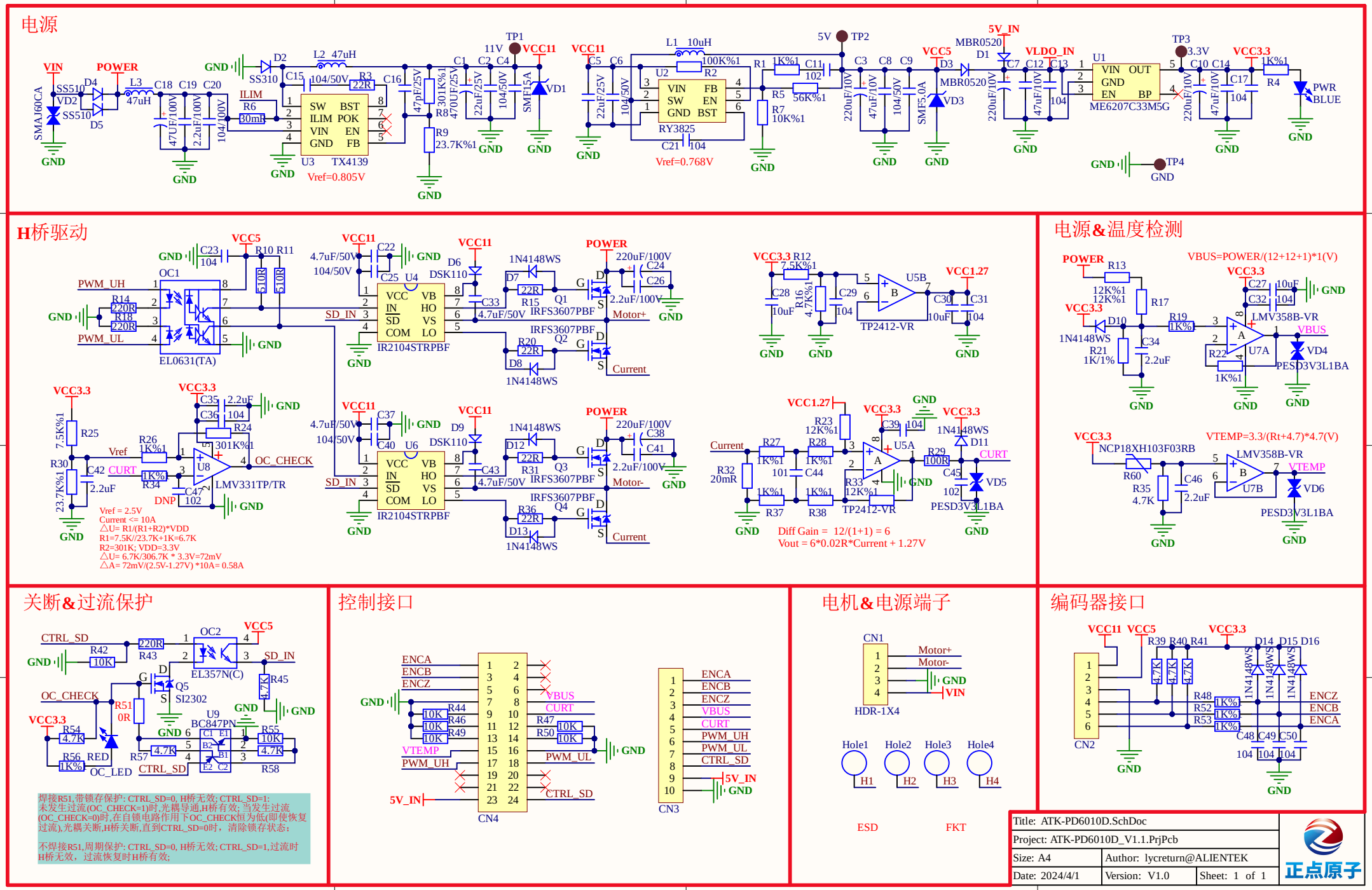
电源
VIN
POWER
SMAJ60CA
SS510
D4
VD2
SS510
D5
L3
47uH
C18
C19
C20
47UF/100V
2.2uF/100V
104/100V
GND
D2
SS310
L2 47uH
ILIM
R6
30mR
C15
104/50V
R3
22R
SW BST
ILIM POK
VIN EN
GND FB
U3 TX4139
Vref=0.805V
C16
47pF/25V
R9
23.7K%1
R8
301K%1
470UF/25V
C1
C2
C4
22uF/25V
104/50V
11V
TP1
VCC11
SMF15A
VD1
L1 10uH
100K%1
R2
U2
VIN FB
SW EN
GND BST
RY3825
C21
104
Vref=0.768V
C5
C6
R1
1K%1
C11
102
5V
TP2
R5
56K%1
R7
10K%1
C3
C8
C9
220uF/10V
47uF/10V
104/50V
VCC5
D3
MBR0520
SMF5.0A
VD3
5V_IN
MBR0520
D1
VLDO_IN
C7
C12
C13
U1
VIN OUT
GND
EN BP
ME6207C33M5G
TP3
3.3V
VCC3.3
C10
C14
C17
1K%1
R4
PWR
BLUE
TP4
H桥驱动
VCC5
R10 R11
510R
510R
OC1
PWM_UH
PWM_UL
R14
220R
R18
220R
EL0631(TA)
C23
104
VCC11
C22
4.7uF/50V
104/50V
C25
U4
VCC
IN
SD
COM
VB
HO
VS
LO
IR2104STRPBF
SD_IN
D6
DSK110
C33
4.7uF/50V
1N4148WS
D7
22R
R15
Q1
IRFS3607PBF
POWER
220uF/100V
C24
C26
2.2uF/100V
Motor+
R20
22R
Q2
D8
1N4148WS
Current
VCC3.3
R25
7.5K%1
R30
23.7K%1
R26
1K%1
Vref
C35
2.2uF
C36
104
R24
301K%1
U8
LMV331TP/TR
OC_CHECK
C42
2.2uF
CURT
R34
1K%
C47
102
DNP
Vref = 2.5V
Current <= 10A
△U= R1/(R1+R2)*VDD
R1=7.5K//23.7K+1K=6.7K
R2=301K; VDD=3.3V
△U= 6.7K/306.7K * 3.3V=72mV
△A= 72mV/(2.5V-1.27V) *10A= 0.58A
C37
C40
U6
D9
D12
R31
Q3
C43
R36
Q4
D13
C38
C41
Motor-
VCC3.3 R12
7.5K%1
C28
10uF
R16
4.7K%1
C29
104
U5B
TP2412-VR
VCC1.27
C30
10uF
C31
104
VCC1.27
R23
12K%1
R27
1K%1
101
C44
R28
1K%1
R32
20mR
R37
1K%1
R38
1K%1
R33
12K%1
U5A
TP2412-VR
C39
104
R29
100R
C45
102
VD5
PESD3V3L1BA
D11
1N4148WS
Diff Gain = 12/(1+1) = 6
Vout = 6*0.02R*Current + 1.27V
电源&温度检测
POWER
R13
12K%1
12K%1
R17
VCC3.3
D10
1N4148WS
R21
1K/1%
C34
2.2uF
R19
1K%
VBUS=POWER/(12+12+1)*1(V)
C27
10uF
C32
104
LMV358B-VR
U7A
R22
1K%1
VD4
PESD3V3L1BA
NCP18XH103F03RB
R60
R35
4.7K
C46
2.2uF
VTEMP=3.3/(Rt+4.7)*4.7(V)
LMV358B-VR
U7B
VTEMP
VD6
PESD3V3L1BA
关断&过流保护
CTRL_SD
R42
10K
R43
220R
OC2
EL357N(C)
SD_IN
R45
4.7K
Q5
SI2302
OC_CHECK
R51
0R
U9
BC847PN
VCC3.3
R54
4.7K
R56
1K%
RED
OC_LED
R57
4.7K
CTRL_SD
R58
4.7K
R55
10K
焊接R51,带锁存保护: CTRL_SD=0, H桥无效; CTRL_SD=1:
未发生过流(OC_CHECK=1)时,光耦导通,H桥有效; 当发生过流
(OC_CHECK=0)时,在自锁电路作用下OC_CHECK恒为低(即使恢复
过流),光耦关断,H桥关断,直到CTRL_SD=0时，清除锁存状态;
不焊接R51,周期保护: CTRL_SD=0, H桥无效; CTRL_SD=1,过流时
H桥无效，过流恢复时H桥有效;
控制接口
ENCA
ENCB
ENCZ
VTEMP
PWM_UH
5V_IN
R44
10K
R46
10K
R49
10K
VBUS
CURT
R47
10K
R50
10K
PWM_UL
CTRL_SD
CN4
CN3
电机&电源端子
CN1
Motor+
Motor-
VIN
HDR-1X4
Hole1
Hole2
Hole3
Hole4
H1
H2
H3
H4
ESD
FKT
编码器接口
VCC11
VCC5
VCC3.3
R39
R40
R41
D14
D15
D16
1N4148WS
R48
R52
R53
1K%
C48
C49
C50
104
CN2
Title: ATK-PD6010D.SchDoc
Project: ATK-PD6010D_V1.1.PrjPcb
Size: A4
Author: lycreturn@ALIENTEK
Date: 2024/4/1
Version: V1.0
Sheet: 1 of 1
正点原子